Final Project

MEC 560: Advanced Control Systems

Spring 2026

Instructor	Amin Fakhari, Ph.D.
Assigned Date	Friday, May 1, 2026
Due Date	Tuesday, May 12, 2026, 11:59 PM

1 Project Overview

In this project, each student will select a dynamic system and perform a comprehensive modeling, analysis, controller design, observer design, and performance evaluation through simulations, using the concepts developed in the course. The project will be evaluated based on creativity in system selection, the depth of analysis, rigor in control and observer design and evaluation, and clarity of presentation. The project is designed to allow students to apply the theoretical concepts learned in class to a practical engineering problem of their choice.

2 System Requirements

You should choose a realistic, non-trivial dynamical system of engineering interest with a minimum of 4 state variables that features *at least* one of the following: unstable equilibrium point, multi-input multi-output (MIMO) behavior, underactuated, and higher-order dynamics. If the system dynamics is nonlinear, it should be Jacobian linearizable about a well-defined equilibrium point or feedback linearizable.

The system should be complex enough to allow for meaningful analysis and control design, but not so complex that it becomes intractable within the scope of this course. You are encouraged to select systems that are relevant to your interests and future career goals, and that can be reasonably modeled and analyzed using the tools covered in this course. Examples of suitable systems include, but are not limited to, aerospace vehicles (quadcopters, satellites, aircraft, rockets), flexible robotic manipulators, inverted pendulum variants (Furuta pendulum, Segway, overhead crane, load-sway system), magnetic levitation systems, electromechanical systems, or biomechanical models. Examples of unacceptable systems are standard mass-spring-damper combinations, basic pendulums, or simple RLC circuits, unless these are part of a larger multi-body system.

3 Project Requirements

Your project must systematically cover the following phases:

Phase I: Modeling and Open-Loop Analysis

- **Derivation**: Derive the dynamic equations of the system from the first principles. Express the system in state-space form, and clearly define your state variables, inputs, outputs, and any disturbances.
- **Equilibria and Stability**: Identify the equilibrium points and analyze the stability of the points.

- **Linearization**: If the system is nonlinear, linearize it about a practically meaningful equilibrium point (preferably the unstable one) to obtain your $\boldsymbol{A}$, $\boldsymbol{B}$, $\boldsymbol{C}$, and $\boldsymbol{D}$ matrices. Depending on the chosen system and the application, you may also consider using feedback linearization techniques to obtain a linearized model.
- **System Analysis**: Analyze input-output stability, controllability and stabilizability, observability and detectability, and the minimal realization of the linearized system.

Phase II: Controller and Observer Design

- **State-Feedback Control**: Design a state-feedback controller to stabilize the system at the (unstable) equilibrium point. Then, design tracking controllers to follow a step and also a time-varying reference trajectory.
- **Observer Design**: Assume not all states are measurable. Design full-order and reduced-order observers to estimate the unmeasured states.
- **Observer-Based Control**: Combine your controllers and observers and demonstrate closed-loop stability.
- **(Optional) Feedback Linearization**: If your system is nonlinear and feedback linearizable, design a controller for the feedback-linearized system. Compare its performance against the linear controllers designed in previous parts when applied to the original nonlinear system.

Note: In all the controllers, you need to achieve a good transient response (e.g., rise time, settling time, overshoot), depending on the application, and zero steady-state error. Moreover, you should justify the gains you choose for your controllers and observers.

Phase III: Computer Simulation and Robustness Evaluation

- **Simulation**: Simulate the closed-loop system with your designed controllers and observers in Python (using Control Systems Library) or MATLAB (using Control System Toolbox). Python is preferred. Evaluate the performance in terms of stability, transient response, and steady-state error for a few different initial errors.
- **Robustness Analysis**: Introduce uncertainties in your system parameters and evaluate the robustness of your controllers and observers under these uncertainties.
- **Challenge**: Introduce a realistic constraint to your simulation, such as actuator saturation (e.g., motor torque limits), external disturbance, or measurement noise, and investigate how your observer-based controller (or feedback-linearized controller) handles these real-world imperfections.

4 Deliverables

- **Detailed Final Report**: The report should be well-organized and clearly written and include all derivations, plots, and analytical discussions, corresponding to each Phase of the project.
- **Code and Simulations**: All well-commented Python or MATLAB files used to generate your results.

Note: If you use Jupyter notebooks, you can include your report there and submit the notebook file (.ipynb) along with a PDF version of the notebook. If you use MATLAB, you can submit .m files along with a PDF report.